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Comparing Return to Play Protocols After Sports Related Concussion Among International Sporting Organisations

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Abstract

Background: Return to play (RTP) protocols are an important part of recovery management following a sport related concussion (SRC) and can prevent athletes from returning to competition too early and thereby avoid prolonged recovery times. To assist sporting organisations in the development of RTP guidelines, the Concussion in Sports Group (CISG) provides scientific based recommendations for the management of SRC in its consensus statement on concussion in sport. **Objectives**: This study investigates commonalities and differences among current RTP protocols of international sporting organisations and examines the implementation of the most recent CISG recommendations. **Methods:** Concussion guidelines and medical rules of 12 international sporting organisations from contact, collision and combat sports were accessed via the organisations websites and compared regarding the management of SRC and the RTP decision. **Results:** Only six of the included organisations developed and published their own concussion guidelines which included a RTP protocol on their website. The number of steps until RTP was similar across the different protocols. Each protocol required at least one medical examination before clearing an athlete to RTP. A high variation among organisations was found for initial resting period after injury, the implementation of sport-specific training drills and the time needed to complete the protocol before returning to competition. At the date of this study (September 09, 2023), none of the accessible RTP protocols were updated to include the latest version of the CISG consensus statement **Conclusion:** To improve the safety of athletes after a head injury, sporting organisations should develop sport-specific guidelines according to the latest CISG consensus statement and this should be updated regularly. Implementation is especially important in combat sports, where there is a high incidence of head injury. Thus, there is a requirement for the most up to date concussion management protocols in these sports.

Keywords: Sport related Concussion, concussion guidelines, concussion management, Return to Play,

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RTP, international sporting organisations

1. Introduction

Despite the improvement of management strategies for sport-related concussion (SRC) over the last years, the decision of when an athlete can return to play (RTP) safely is yet a complex and multidimensional task [1, 2]. Returning too early can lead to prolonged recovery times and increase the risk of sustaining additional injuries such as a subsequent concussion, second impact syndrome or musculoskeletal injuries [3-5]. High heterogeneity in the determination of clinical recovery and days needed for unrestricted RTP in existing literature [6-9] makes it even more complicated for healthcare professionals to make an appropriate decision on when to clear an athlete to RTP, resulting in 43.5% of concussed patients returning to play too soon [10] and while still being symptomatic [6].

Providing optimal recovery strategies, including specific RTP guidelines, is an important part of concussion management and is associated with reduced subsequent concussion rates [8]. To assist healthcare professionals in the management of SRC, the consensus statement of the Concussion in Sport Group (CISG) gives recommendations on how to develop concussion management protocols, including a graduated return to play (RTP) strategy [8]. The RTP strategy consists of 6 steps, which guide an athlete from symptom-limited activity (Step 1) to full return to play (Step 6) using a stepwise approach to increase physical and cognitive activity with each step taking a minimum of 24 hours [8]. Following the international conference on concussion in sport held in Amsterdam in 2022, the CISG revised their previous statement from 2017 [11] and published a new consensus statement that includes modified RTP recommendations. This resulted in an adjusted version of the Sport Concussion Assessment Tool 6 (SCAT6) [12] and the introduction of the new Sport Concussion Office Assessment Tool 6 (SCOAT6) [13] for evaluations 72 hours after injury [8]. While the 2017 version included an initial relative rest period of 24 - 48 hours before starting symptom-limited

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activity, the 2023 version advises athletes to start with the first step within the first 24 hours following injury. This recommendation is congruent with recent research showing that light intensity physical activity (activities of daily living, and sub-symptom threshold aerobic exercise) does not exacerbate symptoms and is beneficial for recovery after SRC, helping in faster RTP [4, 14]. Another modification in the 2023 statement [8] is the addition of moderate aerobic exercise (up to 70% of maxHR) as a sub-step of phase 2, following light aerobic exercise (up to 55% of maxHR). Additionally, athletes may begin light resistance training from step 2, and may continue to exercise even in the case of mild exacerbation of symptoms during steps 1-3 and only need to return to a previous step when symptoms occur during steps 4-6. The start of resistance training was initially only allowed in later steps (3-4 at the earliest) and athletes were advised to stop exercise and return to the previous asymptomatic stage when symptoms occurred at any phase during the RTP program. Another new approach in the 2023 version is that athletes may only begin step 4 of non-contact training drills when they are authorized by a healthcare professional after resolution of cognitive impairments and other symptoms related to concussion at rest, during or after physical exercise. Earlier versions of RTP strategies focused mostly on the absence of concussion symptoms, while cognitive testing was underutilized in the decision making [5]. However, ensuring that athletes are free from cognitive impairments is an important aspect of the RTP decision. When comparing the performance in a switch task, to non-concussed controls, a high number of concussed athletes showed a poorer response accuracy indicating an incomplete cognitive recovery, although they were cleared to RTP by a healthcare professional [5].

The revised return to play strategy provided by the CISG provides a science-based summary of evidence and gives general practice recommendations for the management of SRC. However, the recommendations are 'not intended as a clinical practice directive or legal standard of care and should not be interpreted as such' [8]. Therefore, to ensure that their athletes return to play only after fully complete recovery, sports associations and governing bodies should provide appropriate gradual return to play protocols [15] based on the consensus statement. However, current concussion guidelines and RTP recommendations among both national and international sporting associations

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show high inconsistency and a lack of detail [15-17]. In a comparison of sport-specific concussion guidelines among 15 sporting associations in the United Kingdom, Scullion & Heron [15] found large discrepancies in the RTP strategies, with the recommended minimum time before returning to play ranging between 6 and 35 days. The authors provide details on initial resting times, the implementation of medical clearance before RTP, and how to handle athletes with a history of multiple concussions, but important information such as the number of included steps before RTP or the possibility of completing an advanced RTP program are absent. Also, the review lacks information on protocol contents such as when to implement resistance training and the existence of sport-specific training recommendations. In addition, inclusion criteria for sporting organisations were based not only on the sports risk of sustaining a concussion, but also on the participation level within the United Kingdom. As a result, sports with high risk of sustaining a concussion, for example karate, taekwondo, or kickboxing, were not included in Scullion & Herons review. Likewise, in another review of RTP protocols of sporting organisations in Ireland, Beakey et al. [16] claim that there exists considerable variation from CISG guidelines and that recommendations on concussion management are lacking in details and clarity, which might lead to ambiguity among athletes. The review includes details on the minimum time until RTP, initial resting periods, and how to handle athletes with prolonged symptoms, but again information on the number of RTP steps, the implementation of resistance training and the inclusion of sport-specific training drills was not included. In other research including both national and international associations, Davis et al. [17] report a high variety in the implementation of CISG recommendations among existing SRC guidelines. When investigating variations from the CISG recommendations, the authors mention the existence of football-specific training skills implemented in the RTP guidelines provided by the Football Association (England) and the possibility for an advanced RTP pathway in the guidelines provides by the National Rugby League (Australia). The authors also mention prolonged suspension times for athletes with consecutive concussions within 12 months which were implemented in the International Boxing Associations RTP protocol. However, no detailed information is given concerning the total time until RTP, initial resting time or number of RTP steps.

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In summary, the existing literature on RTP guidelines provided by both national and international sporting organisations provides important information regarding differences in the total time until RTP or the duration of initial resting periods, but the inclusion of protocol-specific contents like the implementation of resistance training, the number of included RTP steps, and sport-specific training recommendations need further investigation for clarity. Following the revision of CISG recommendations in 2023, the adherence of existing RTP protocols among international sporting organisations to the latest version of the consensus statement needs to be reinvestigated as well. Therefore, the purpose of this commentary was to compare current return to play strategies by international sporting organisations of combat, contact and collision sports in which athletes are at a high risk of sustaining a concussion. International sporting organisations serve as the governing body for athletes around the world and therefore need to provide clear and consistent RTP guidelines to maintain and improve their athlete's safety. Besides investigating commonalities and differences among federations, this commentary will also discuss the implementation of current CISG recommendations following the latest revision of the consensus statement on concussion. Through this investigation we aim to ascertain the potential inconsistencies among current RTP protocols and provide suggestions to improve future concussion guidelines to maximize safety prior to returning to play.

2. Materials and Methods

The main research questions for this commentary were (1) Which sporting organisations do provide specific RTP protocols or recommendations for the management of SRC? (2) What are the commonalities and differences among protocols and recommendations for different sports? (3) What are the differences between current guidelines provided by international sporting organisations and the revised recommendations by the CISG? To answer these questions, current concussion guidelines from international sporting organisations were examined based on their recommendations for both the management of SRC and the return-to-play decision.

2.1. Eligibility criteria

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Concussion guidelines or RTP protocols for combat sports, contact sports and collision sports were eligible for this study, due to the high risk of sustaining a sports-related-concussion [11]. To be considered for inclusion, the sport had to be part of the Olympic program or had to be governed by an international organisation recognized by the International Olympic Committee (IOC). Only organisations providing official concussion guidelines or a return-to-play protocol available in English were included. In the absence of concussion guidelines or a RTP protocol, an organisation was still included, if the medical rules provided recommendations regarding the management of SRC or the suspension from competition. Organisations that neither provided concussion guidelines nor recommendations within the medical rules were excluded. If more than one international organisation existed for the same sport, the organisation recognized by the IOC was selected. For boxing, the organisation last recognized by the IOC was included even though it was barred in 2022.

2.2. Search Strategy

Consistent to existing research on RTP guidelines among sporting organisations [15-17], the most current guidelines were accessed through the official website of each organisation (date of access: September 09, 2023). Access to each organisation's website was gained through the link provided on the official website of the International Olympic Committee. For organisations that did not provide specific protocols on their website, search for guidelines was performed using a web browser (Google). Keywords used to search for eligible protocols included "concussion guidelines", "concussion protocol", "RTP guidelines" or "RTP protocol" in combination with the name of the governing organisation or the specific sport. Protocols retrieved via web browser were only included if they were designated as guidelines published by the official governing body of the corresponding sport. If more than one protocol was available via a web search, the most updated version was included. Medical rules for each organisation were accessed likewise, using the keywords "medical rules", "medical code", "medicine manual", or "medical handbook" in combination with the organisations name. Available documents were then screened using the keywords "concussion", "head injury" and or "loss of consciousness". Associations that did not provide specific guidelines were

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contacted via their official mail address on the website to ensure the absence of corresponding documents.

2.3. Data extraction

Available documents (concussion guidelines, RTP protocols, medical rules) for each of the included sporting organisations were screened and relevant information was summarized via Microsoft Excel. Considering the research questions, the contents of existing protocols were compared using the following categories: initial resting phases, number of steps included in each protocol, the existence of sport-specific training recommendations, the implementation of resistance training within the protocol, need of medical clearance before RTP, the minimum time needed for RTP, and the existence of accelerated or advanced RTP protocols.

3. Results

14 sporting organisations were considered eligible for inclusion and their official websites were screened for the availability of concussion guidelines and recommendations within or separate to the medical rules. Due to the absence of applicable documents, two of these organisations had to be excluded from further investigation. Therefore, a total of 12 different sporting organisations, including the Federation Internationale de Basketball (FIBA) [18], Federation Internationale de Football Association (FIFA) [19, 20], Federation Internationale de Hockey (FIH) [21], International Judo Federation (IJF) [22], International Boxing Association (IBA) [23], International Federation of American Football (IFAF) [24, 25], International Ice Hockey Federation (IIHF) [26, 27], World Association of Kickboxing Organisations (WAKO) [28], World Karate Federation (WKF) [29], World Lacrosse (WL) [30, 31] World Rugby (WR) [32, 33] and World Taekwondo (WT) [34] were included and their guidelines and medical rules assessed.

3.1. Existence of RTP guidelines and medical rules

Table 1 gives an overview of published guidelines and medical rules by sporting organisation. Six out of the 12 included organisations (FIBA, FIFA, IBA, IFAF, IIHF, WR) [18, 19, 20, 23, 24, 25, 26, 27,

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32, 33] have published their own concussion guidelines with an included RTP protocol and provided additional recommendations on concussion and head injury in their medical rules. For FIBA and IBA, concussion guidelines and the RTP protocol were included as part of the provided medical rules. WL did provide a concussion policy on their website, but the guidelines did not include a specific RTP protocol. Instead, they uploaded the Sport Concussion Assessment Tool 5 (SCAT5) [21], which includes RTP recommendations provided by the CISG, as supplementary document on their website. WT and the FIH did not publish sport-specific guidelines but also uploaded the SCAT5 [21] on their website. While WT also included post-injury recommendations [34] and suspension rules in their medical code, no concussion management advice was found in the medical guidance provided by the FIH. Three federations (IJF, WAKO, WKF) [22, 28, 29] provided no specific RTP guidelines but included recommendations following head injury and concussion in their medical rules.

Table 1. Concussion guidelines and medical rules by organisation

Organisation	***Concussion Guidelines***	***Medical Rules***
Federation Internationale de Basketball (FIBA)	Basketball Concussion Management Guidelines	Medical Resource for Basketball Team Physicians
Federation Internationale de Football Association (FIFA)	FIFA Medical Concussion Protocol	Football Emergency Medicine Manual 2nd Edition
Federation Internationale de Hockey (FIH)	SCAT5	NA*
International Judo Federation (IJF)	NA	Sport and Organisation Rules
International Boxing Association (IBA)	Graduated Return to Boxing Program	IBA Technical & Competition Rule
International Federation of American Football (IFAF)	IFAF Concussion	Game Day Medical Arrangements – Tackle Football
International Ice Hockey Federation (IIHF)	IIHF Concussion Protocol	IIHF Medical Regulations
World Association of Kickboxing Organisations	NA	Medical Rules

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(WAKO)		
World Karate Federation (WKF)	NA	Medical Rules
World Lacrosse (WL)	World Lacrosse Concussion Policy: Identification, Treatment and Management + SCAT5	World Lacrosse Handbook
World Rugby (WR)	World Rugby Concussion guidance	World Rugby Medical Regulations and Laws
World Taekwondo (WT)	SCAT5	WT Medical Code

*A 'Medical Officers Manual' is available on the FIH website but was not included due to missing concussion information.

3.2. Initial resting phase

Initial resting periods before starting physical activity are included in all existing concussion guidelines, except the one from the IFAF (Table 2). A time frame of 24 to 48 hours of cognitive and physical rest is requested by most organisations (FIBA, FIFA, FIH, IIHF, WL, WT) [18, 19, 20, 21, 26, 30] before starting physical activity, while stricter resting phases are required for Rugby (WR) [32] and Boxing (IBA) [23]. The only organisation recommending different resting periods for adult and child athletes was WR [32]. While adults must complete one week of relative rest (including 24 hours of complete rest), children must wait for two weeks (including 24 hours of complete rest) before starting the RTP protocol. In boxing athletes must complete a suspension period of 30 days before advancing to step 2 of the RTP program, although the IBA allows starting symptom-limited activity (step 1) after an initial period of 24 to 48 hours complete rest. The IFAF [24] does not report a defined time frame for resting but requires an athlete to be free of symptoms and cleared by a medical before starting the RTP protocol. No clear instructions on resting periods before starting physical activity following concussion were found for IJF, WAKO and WKF [22, 28, 29].

3.3. Number of RTP steps

Eight organisations require a total of six steps leading an athlete to a safe return to play (FIFA, FIH,

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IBA, IIHF, WL, WR, WT) [19, 20, 23, 26, 27, 30, 32]. Only the FIBA [18] varied from this recommendation with their protocol consisting of only five steps (Table 2). Regarding the duration of each step, all existing concussion protocols demand each step lasts a minimum of 24 hours before advancing to the next step. In case of returning or worsening of symptoms, athletes must return to the previous step in all the guidelines.

3.4. Sport-specific training recommendations

Individualized RTP protocols with sport-specific training contents and recommendations were provided by the FIBA [18], FIFA [19] and IBA [23]. While FIBA and IBA kept sport-specific training recommendations very simple, the Graduated Return-to-Football Programme for high-level players (FIFA) included detailed football-specific drills for each step. Other existing guidelines provided only general recommendations given in the CISG statement and the SCAT5.

3.5. Resistance Training

Recommendations on when to implement resistance training in the RTP process vary among organisations (Table 2). Both FIFA [19] and IFAF [24] allow athletes to start resistance training simultaneous with moderate activity and sport-specific exercise (step 3). For FIH, IBA, IIHF, WL, WR and WT [21, 23, 26, 30, 32], the start of progressive resistance training is included in step 4 when starting with non-contact training skills at high intensity. No instructions for the implementation of resistance training are given in the FIBA [18] concussion guidelines.

Table 2. Initial resting phase (hours), number of steps in RTP program and implementation of resistance training by organisation

Organisation	***Initial resting phase (hours)***	***RTP steps***	***Sport-specific RTP recommendations***	***Resistance training***
FIBA	24h (relative rest)	5	Yes	NA
FIFA	24 – 48h (cognitive + physical rest)	6	Yes	From step 3
FIH	24 – 48h (cognitive + physical rest)	6	No	From step 4
IJF	NA	NA	No	NA

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IBA	24 – 48h (complete rest)	6	Yes	From step 4
IFAF	NA	6	No	From step 3
IIHF	24 – 48h (cognitive + physical rest)	6	No	From step 4
WAKO	NA	NA	No	NA
WKF	NA	NA	No	NA
WL	24 – 48h (cognitive + physical rest)	6	No	From step 4
WR	Adults: 168h (24h complete rest) Children: 336h (24h complete rest)	6	No	From step 4
WT	24 - 48h (cognitive + physical rest)	6	No	From step 4

3.6. Medical Clearance

All 12 sporting organisations include at least one medical examination in the process of returning to play, with medical clearance being mandatory before finally returning to competition. Before an athlete is allowed to start the RTP program medical clearance is required by FIFA, IFAF and WR [19, 24, 32]. Following SCAT5 recommendations FIFA, FIH, IBA, IIHF, WL, WR and WT [19, 21, 23, 26, 30, 32] programs require clearance by a healthcare professional before starting full-contact training. FIBA guidelines also include medical examination before starting training, but a specific stage of training is not defined.

3.7. Time to complete RTP

The minimum amount of time needed before an athlete is allowed to return to sport was calculated by adding the initial resting period before starting activity to the minimal time frame required to complete each step of the RTP program. If the medical rules of an organisation included a suspension period after sustaining a concussion, the suspension period was defined as time needed to RTP. Among the different associations, the time before an athlete is allowed to return to play had a high variation, ranging from five days to 180 days from the occurrence of a single concussion. The amount of time needed to complete the RTP was shortest in FIBA [18] guidelines and longest in the WAKO

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medical rules [28]. Results for minimal time to complete the RTP protocol are shown in Table 3.

3.8. Fast RTP

The possibility to complete an advanced or accelerated RTP protocol and return to competition faster is given in the guidelines of FIFA, IIHF and WR [19, 26, 32]. No advanced or accelerated RTP programs were found for the remaining sporting organisations.

3.9. Compliance with newest CISG recommendations

At the date of this commentary (September 09, 2023) none of the accessible RTP protocols was updated to the newest findings of the 2023 CISG consensus statement [8]. All guidelines were still based on the recommendations from the 2017 CISG consensus statement [11].

Table 3. Number of medical examinations within the RTP program and minimum time until RTP

Organisation	*Medical clearance*	*Time to complete RTP (days)*
FIBA	Before return to training, Before return to play	5 days
FIFA	Before starting RTP protocol, Before starting contact training, Before return to play	7 days
FIH	Before starting contact training, Before return to play	7 days
IJF	Before return to play	8 days
IBA	Before starting contact training, Before return to play	35 days, Second concussion: 35 days, Third concussion: 365 days
IFAF	Before starting RTP protocol, Before return to play	6 days
IIHF	Before starting contact training, Before return to play	7 days
WAKO	Before return to play	30 days (no LOC), 90 days (LOC < 1 min), 180 days (LOC > 1 min), Second concussion: 90 days

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		Third concussion: 365 days
WKF	Before return to play	14 days, Second concussion: 90 days, Third concussion: 365 days
WL	Before starting contact training, Before return to play	7 days
WR	Before starting RTP protocol, Before starting contact training, Before return to play	Adults: 14 days Children: 21 days
WT	Before starting contact training, Before return to play	Senior: 30 days, Junior: 40 days, Cadet: 50 days, Second concussion: 90 days, Third concussion: 180 days

Abbreviations: LOC – Loss of consciousness.

4. Discussion

To our knowledge, this is the first study to compare concussion guidelines and medical rules of international sporting organisations with a focus on the RTP decision making process. Previous research included only national organisations or compared guidelines in a mix of national and international associations and, although information on recovery and RTP is included, important details like the number of RTP steps per protocol, the implementation of sport-specific training recommendations and the implementation of resistance training guidelines are missing. Therefore, this commentary on current RTP protocols and medical rules provides a detailed comparison of SRC recovery strategies and RTP protocols among 12 international sporting organisations which have a high incidence of concussion. The first research question in this study examined which sporting organisations provide specific RTP protocols to guide athletes in the recovery from SRC. Of the 12 included organisations only 6 (FIBA, FIFA, IBA, IFAF, IIHF, WR) provided specific designed RTP protocols. Three organisations (FIH, WL, WT) uploaded the SCAT5 tool that includes recommendations for the RTP process but did not publish their own recommendations. Three organisations (IJF, WAKO, WKF) provided suspension periods following head injury, but did not

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publish specific RTP recommendations.

RTP protocols should provide guidance for athletes to return to physical activity and competitive play safely and prevent them from returning to sports too early. To increase athletes' safety and provide useful recommendations, protocols should be up to date, match current scientific findings and expert recommendations, and should be consistent with other existing guidelines to avoid ambiguity. Thus, the second research question focused on commonalities and differences among the protocols of different sports. With the exception of the protocols from IFAF and WR, all available protocols included an initial resting period lasting 24-48 hours following concussion. The number of total RTP steps before returning to competition was the same for all available protocols apart from the FIBA protocol. All available protocols included medical clearance before finally returning to competition, while differences in medical clearance before starting the RTP program and before starting contact training exist. A high inconsistency was found regarding the implementation of resistance training, the existence of sport specific training recommendations and the total time needed to complete the RTP program. Considering that physical activity can be beneficial for recovery and accelerate RTP [4, 14], protocols should consistently follow current research recommendations. Although time to RTP might differ among individuals [6-9], a minimum time before an athlete is allowed to return to play should be defined according to current scientific findings and should be consistent throughout different sports. Considering that non-athletes, in particular non-professional or recreational sporting participants may be active in more than one sport, inconsistent recommendations following injury can confuse athletes in the RTP progress.

To improve consistency among RTP protocols, the consensus statement published by the CISG can function as a framework for each organisation. The consensus statement is based on newest scientific findings and its fundamental steps and contents should be followed by governing bodies of sport. Following the publication of the newest version in June 2023, our third question investigated whether sporting organisations have implemented the revised recommendations in their medical rules and guidelines. Unfortunately, at the date of this study (September 09, 2023), none of the available

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RTP protocols was updated with the newer recommendations. As mentioned earlier in this commentary, the revised version of the CISG statement includes important modifications and not following these modifications may be a potential source of prolonged recovery times or even subsequent injury.

4.1. Existence of RTP guidelines

As mentioned, the number of existing concussion guidelines with an included RTP strategy was alarmingly low, with only six of the 12 organisations providing specific designed RTP protocols for their athletes. Additionally, considering that two organisations were excluded from this study due to the lack of any concussion related information on their websites, this means that less than half of the sporting organisations provide sufficient guidance to ensure their athletes safety. Although FIH, WL and WT provided the SCAT5 [21], the CISG declares that their consensus statement is not meant to serve as legal standard but should rather be viewed as general advice to develop individual protocols and recommendations [8]. In the newest statement, they also state that the clinical utility of previous versions of the SCAT diminishes after 72 hours and that they might not be the best choice for making RTP decisions beyond 7 days postinjury [8]. To promote athletes' safety, the IOC as authority organising the Olympic Games should implement the existence of appropriate concussion rules as a requirement for organisations to be officially recognized by the IOC.

Concurrent with previous research [15, 16], the accessibility of guidelines was low as the search for concussion guidelines and medical rules was time consuming. Protocols were difficult to find on the organisation's websites and required additional web browser searches. Additionally, requests for supplementary documents and information were not answered by any of the organisations contacted via mail. It is especially important for athletes and teams with limited access to professional healthcare that RTP protocols are easy to access. To guarantee accessibility and to minimize potential health complications, sporting organisations need to implement detailed and appropriate RTP protocols that allow for effective SRC management and improved safety of athletes [8, 11, 35]

When focusing on martial arts, we found that the existing recommendations following

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concussion or head injury are very poor in extent and detail, especially for Judo, Kickboxing and Karate. IJF, WAKO and WKF [22, 28, 29] did not provide specific concussion guidelines and the recommendations within the medical rules are mostly limited to suspension periods following knock out or head injury. Considering that martial arts include kicking and punching techniques to the opponents' head and throwing techniques that can create fast acceleration of the opponents' body and head, athletes are at high risk of sustaining head injury and therefore need appropriate regulations to improve their safety.

4.2. Initial resting phase

Including a period of relative rest of 24-48 hours is consistent with CISG recommendations [8], however the definition of the required rest is different among organisations. While the FIBA [18] recommends a phase of 24 hours of relative rest, the Graduated Return to Boxing Program (IBA) [23] mandates a period of 24 to 48 hours of complete physical and mental rest before athletes may start with daily activities. In Rugby (WR) [32], the resting phase of one to two weeks includes a period of complete physical and cognitive rest within the first 24 hours. The guidelines of FIFA, FIH, IIHF, WL, and WT [19, 21, 26, 30] include a period of brief physical and cognitive rest but do not define whether relative or complete rest is required.

In their newest consensus statement, the CISG advises athletes to start with step 1 of the RTP protocol during the initial resting period within 24 hours following injury instead of completing a phase of absolute rest [8]. Research shows that strict rest is counterproductive, and that appropriate activity may facilitate recovering from a concussion [4, 14, 36, 37]. Considering this advice, organisations should revise and specify their recommendations, allowing athletes to start activities of daily living within the first 24 hours of injury. Complete resting phases as prescribed in Rugby and Boxing should be reconsidered and updated.

4.3. Number of RTP steps

The number of included steps until full RTP were congruent with the current strategy provided by the

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CISG [8]. According to the newest version of the consensus statement, the recommended content for each step of the RTP protocol are as follows: symptom-limited activity (step 1), aerobic exercise (step 2), individual sport specific exercise (step 3), non-contact training drills (step 4), full contact practice (step 5) and return to play (step 6). Compliance with this exercise strategy was found in FIH, IBA, IFAF, IIHF, WL, WR, and WT [21, 23, 24, 26, 30]. FIFA [19] also followed CISG recommendations but included an additional sub-stage (step 4.2) of controlled contact drills before starting with full-contact practice.

In the FIBA guidelines [18], symptom-limited activities are not included, and athletes may start with light aerobic exercise and easy basketball skills (step 2) following a relative rest period (step 1). Then, after completing step 3 (light non-contact training), players are allowed to participate in scrimmage (step 4) after medical clearance and then return to play (step 5). Unfortunately, the description of each step is not very specific and there is no detailed information on allowed activities during the first step. To enhance the protocol to be better understood, the contents of each step need a clear description that athletes can follow and implement to their training regime.

4.4. Sport-specific training recommendations

The consistency of the basic contents of the RTP recommendations of sporting organisations is important to avoid confusion and improve safety. Therefore, the framework of provided protocols (i.e. initial resting time, minimum time to complete RTP protocol, need for medical clearance) should be based on current scientific evaluations and must be consistent among all sporting organisations. Nevertheless, providing additional training recommendations and training contents for each specific sport and for each step within the protocol may help athletes in completing the protocol and return to their performance level prior to injury. In elite sports, athletes are supervised by medicals as they progress through the RTP program and the additional support by an athletic trainer may help in restoring function and performance [38]. However, at a non-elite level RTP protocols are usually completed without the support of healthcare professionals [15] and progressing properly through each of the steps might be difficult. Therefore, detailed training recommendations provided by the

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organisation may be beneficial for guiding both professional and non-professional athletes through each step and preparing them to participate in competition.

Both FIBA [18] and IBA [23] give brief examples such as “free throws and shooting” for light aerobic exercise in basketball and “Bag/Pad work” for non-contact drills in boxing which may help athletes understand what kind of drills they are allowed to do at a certain point of the RTP protocol. Nevertheless, these explanations lack detail and important information such as training intensity is missing. Providing more details on these recommendations should be a goal for future improvements of RTP protocols provided by FIBA and IBA.

More details and sport-specific recommendations are provided by the FIFA [19], including examples on training contents and information on intensity while performing these exercises. To improve athlete health care, other organisations should follow and provide more specific protocols that help guide athletes or teams without professional health care staff.

4.5. Resistance Training

Recommendations on when to implement resistance training among the assessed protocols are based on the 2017 CISG statement [11], allowing resistance training earliest from step 3 or step 4. According to the newest statement from 2023, athletes may start with light resistance training from step 2 if it does not lead to more than mild worsening of symptoms [8].

Although resistance training may provide an alternative way to reintroduce exercise following a concussion [36], it is not well established as a common form of training used in concussion management due to concerns including increased intracranial pressure, distinct changes in blood pressure and possible exacerbation of symptoms [36, 39]. In a feasibility study Hutchinson et al. [36] introduced a continuous aerobic resistance exercise protocol including a combination of bodyweight resistance exercises (i.e. air squats, shoulder press) with musculoskeletal and cardiovascular demands. Although the authors reported no adverse effects following the protocol and

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participants showed a reduced symptom burden, the validity of these results remains unclear. To our knowledge, there is yet no research that investigates the safety and effectiveness of resistance training including weights in the early stages of RTP and its outcomes on concussion recovery. Unfortunately, the 2022 CISG statement does not include scientific evidence or an explanation for changes made in the prescription of resistance training compared to the 2016 statement. Therefore, more research is needed to evaluate possible benefits from resistance training, before RTP protocols can adapt this revision.

4.6. Medical clearance

Each of the included organisations required at least one mandatory medical examination before allowing an athlete to return to competition. This is important to improve the safety of athletes and to prohibit premature RTP. However, we found a high variation in the number of required medical examinations as well as in the stage at which the examination needs to be performed. According to the newest CISG statement [8], athletes should be authorized by a health care professional before starting activities at risk of collision or contact (steps 4-6). If sport-specific activity (step 3) includes the risk of head impact, athletes should also be medically cleared prior to starting that step. The guidelines specify that a medical examination at this point of the RTP protocol should assure the resolution of existing symptoms and the absence of cognitive impairments at rest as well as after physical exertion. Additionally, a medical or health care professional should give written confirmation to attest to the readiness for RTP before an athlete can return to competition unrestricted.

Organisations should follow the new approaches and include medical examinations prior to the early stages of training if they include the risk of contact and collision. Including the assessment of cognitive readiness is also an important addition in the newest recommendations especially considering the low reliability of symptom reports [5]. Using cognitive testing in combination with symptom assessment is therefore more sensitive and should be a part of reliable RTP protocols [5].

4.7. Time to complete RTP

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The CISG statement declares that unrestricted RTP may take up to one month and athletes should expect at least one week to complete a RTP program [8]. According to recent research on clinical recovery from concussion, it takes athletes 19.8 to 21 days to return to competition without restrictions [7, 9, 40]. The calculated minimal recovery time from the assessed RTP protocols was shortest in basketball [18]. Although the FIBA states that RTP usually takes around seven to ten days, their RTP protocol consists of five steps (including initial rest as step 1) that can be completed after a minimum of five days if each step is completed over 24 hours. Due to the absence of clear recommendations for initial rest, the RTP strategy provided by the IFAF [24] can be completed after 6 days. As mentioned by Scullion & Heron [15], who found very short recovery period recommendations for Hockey and Ice Hockey organisations in England, allowing athletes to RTP earlier than the recommended period of one week by the CISG may put their health at risk. FIBA and IFAF recommendations should therefore be critically reconsidered and prolonged in recovery time to prevent athletes from returning too early.

Recommendations provided by the remaining organisations stated at least seven days was required to full recovery and are therefore congruent with current CISG advice [8]. The longest restriction period was found in the WAKO medical rules [28], suspending an athlete from training for 180 days in the case of a concussion with loss of consciousness of more than one minute. In general, suspension periods defined in martial arts [23, 28, 29, 34] were longer than the recommended time of 1 week to complete RTP, except for IJF [22]. In case of a subsequent concussion injury, IBA, WAKO, WKF and WT [23, 28, 29, 34] defined stricter suspension rules than for a single occasion, prolonging the time until RTP. A history of concussion is considered as a risk factor for prolonged recovery times and increases the risk of permanent cognitive damage [4, 41]. Extending recovery times due to a history of previous concussions seems appropriate and should be implemented by other organisations and sports as well.

In the WR guidelines [32, 33], differences are made between age groups with children (3 weeks) taking longer time to complete RTP than adults (2 weeks) due to the difference in the initial

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resting period. Similarly, WT [34] distinguishes suspension periods by age-group with cadets (50 days) being excluded from competition longer than juniors (40 days) and seniors (30 days). Remaining organisations did not differentiate between age groups. Compared to adults, children are more vulnerable to concussion and possible long-term consequences due to an immature central nervous system, different volumes of cerebral blood and larger a subarachnoid space [41, 42]. As a result, the mean time required for total RTP is also influenced by age, with younger athletes taking longer to recover [6, 9, 40]. Despite the increased risk in younger age, common guidelines for concussion management are focused on adults with a lack of specific protocols designed for children [41, 43]. While the fifth version of the Sport Concussion Assessment Tool for children from 5 to 12 years [44] includes a RTP protocol that is identical to the one provided for adult athletes in the SCAT5 [21], the newest version of the assessment tool [45] does not provide a specific RTP program for children. In addition, DeMatteo et al. [46] found little evidence on the effectiveness of common RTP protocols in children and states that more research is needed. Considering these results, specific RTP guidelines for children and adolescents need to be designed and evaluated in further research.

4.8. Fast RTP

According to FIFA rules [19], an accelerated Return-to-Football program, focusing on step 2 (aerobic exercise) and step 5 (full-contact training), can be considered if post-injury symptoms are classified as not concussion specific, last for less than 24 hours and a re-evaluation shows normal results. Persisting signs and symptom based orange and red flags, as defined in the concussion guidelines, disqualifies a player of accelerated RTP.

In rugby [32], adult players can avoid the required resting period of one week by accessing an advanced level of concussion care and receiving medical clearance advising that the rest period is not necessary. An advanced care setting thereby includes access to experienced medical doctors, neuroradiologists and brain imaging facilities and a team of multidisciplinary specialists (i.e., neuropsychologist, neurologist). If advanced care is available, the athlete still must complete 24 hours of complete rest, be symptom free and needs medical clearance to RTP.

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In Ice Hockey, contrary to the existing concussion protocol [26], the IIHF medical regulations [27] quote that the RTP decision should be based on individual circumstances rather than on a mandatory period of time. Thus, a player may return to play the day after injury if he/she shows no symptoms at rest and no exacerbation of symptoms at activity levels required to compete and a medical clearance that he/she has returned to his/her neurocognitive baseline.

The requirements for athletes to be allowed to participate in a fast RTP program rely mainly on the recovery of concussion-related symptoms and must therefore be handled with caution [5]. Using only symptom based reports shows low reliability and should therefore not be used exclusively but rather in combination with other assessments such as cognitive testing [5]. Also, athletes might underestimate or downplay existing symptoms to avoid being excluded from competition and speed up RTP [5, 47-49], especially in professional sports [49]. An investigation of RTP behaviour in elite football in the UK [50] showed that the accelerated RTP protocol is used in 66% of concussed athletes, with an even higher percentage (82%) of male first team players using this pathway exclusively. Even more concerning is the fact that in the same study 77% of players used the accelerated RTP without being tested at baseline, even though FIFA clearly defines it as a minimum requirement for enhanced RTP [19]. By hiding or downplaying symptoms players take a high risk and thus the allowance to participate in an accelerated RTP protocols should be considered critically.

4.9. Compliance with CISG recommendations

Although concussion guidelines and RTP protocols provided by the included organisations showed compliance with former CISG recommendations, none of the accessible documents was updated to the 2023 version of the concussion consensus statement at the time of conducting this study. Updates on initial resting times, symptom occurrence and the use of cognitive testing before graduating to later steps (step 4-6) of the protocol were not implemented in any of the accessible guidelines and protocols.

Initial resting time should not include complete physical and cognitive rest as recommended by IBA [23] and WR [32], and athletes should be allowed to begin with the first step within the first

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24 hours after injury to avoid prolonged recovery times [4, 14, 51]. When starting with physical activity, former statements recommended stopping exercise and returning to an earlier stage of the RTP protocol if symptoms occur [11] while the revised statement [8] allows athletes to progress with the protocol if they show only mild symptoms or brief symptom exacerbation (increase of no more than 2 points on a 0–10 point scale for less than an hour when compared with the baseline value reported prior to physical activity). According to current research allowing athletes to progress despite symptom exacerbation does not increase but might even be associated with milder symptoms 2 weeks post-injury and is therefore not harmful [52]. Updated RTP protocols should implement this statement and allow athletes to graduate in the protocol even following the occurrence of mild and brief symptoms. Finally, before allowing an athlete to start with activities at higher risk of impact, monitoring of cognitive function can help make reliable decisions on whether an athlete is ready for such activities [5] and should be included in updated RTP protocols.

To provide evidence-based information according to the newest scientific finding, sporting organisations need to update their concussion guidelines and RTP protocols continually [16]. As mentioned in the consensus statement, the management of SRC will keep evolving in the next years and scientific knowledge will need to be updated to reflect these new findings [8]. International sporting organisations therefore need to implement revised recommendations to improve the safety of their athletes.

5. Conclusion

With the latest revision of their consensus statement, the CISG recommendations provide the latest scientific findings regarding the management of SRC and the RTP process. However, a high number of international sporting organisations do not yet provide specific RTP protocols and existing documents are not updated to include the best evidence available. To improve the safety of their athletes, we recommend that sporting organisations develop sport-specific guidelines and ensure that they are updated regularly to reflect the latest scientific findings. As physical and cognitive requirements to participate in play differ among sports, a single protocol for all sports might not be

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the optimal solution, but fundamentals included in each protocol, like initial resting time, number of RTP steps, the need for medical clearance, and the minimum time to complete the protocol should be consistent among organisations. Additionally providing step wise and sport-specific training recommendations (i.e. training content, intensity) for each organisation may improve adherence and effectiveness of RTP protocols and thereby improve athletes' safety. To further improve safety and achieve best possible outcomes, the benefits of resistance training on concussion outcomes and recovery needs to be investigated.

Conflict of Interest

None of the authors of this study report any conflicts of interest.

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